1 What excess Gaussian energy does and does not control

Write a width-N, depth-L tanh network as $h^1 = \tanh(W^1\bar{x})$, $h^s = \tanh(W^s h^{s-1})$ for $2 \le s < L$, and $z = v^T h^{L-1}$, where $\bar{x} = (1, x)$ and $x \in \{-1, 1\}^d$. Its weighted Gaussian energy is

$$\kappa_N = \frac{d}{N}\|W^1\|_F^2 + \sum_{s=2}^{L-1} \|W^s\|_F^2 + N\|v\|_2^2.$$

The following facts concern this actual energy. They do not establish a sample-complexity lower bound for the unrestricted Gaussian posterior.

Proposition 1.1 (Excess energy controls total deviation from rank one)**.** *Suppose $|z(x)| \ge 1/2$ on the whole Boolean cube. Let R_s be a best rank-one Frobenius approximation to W^s, and let $\rho_s = \|W^s\|_F^2/\|W^s\|_{\mathrm{op}}^2$ be its stable rank. Then*

$$\sum_{s=2}^{L-1} \|W^s - R_s\|_F^2 \le \kappa_N - L(d/4)^{1/L}, \tag{1}$$

$$\sum_{s=2}^{L-1} \log \rho_s \le L \log(\kappa_N/L) - \log(d/4). \tag{2}$$

In particular, if $\kappa_N \le L + B$, both right-hand sides are at most $B - \log(d/4)$.

Proof. Set $a = \|W^1\|_F/\sqrt{N}$, $c = \sqrt{N}\|v\|_2$, and $p_s = \|W^s\|_{\mathrm{op}}$. Cube averaging supplies an input with $\|W^1\bar{x}\|_2/\sqrt{N} \le a$. Tanh contraction gives $1/2 \le ac\prod_s p_s$. Thus the product of the L numbers $da^2, p_2^2, \ldots, p_{L-1}^2, c^2$ is at least $d/4$, and their sum is at least $L(d/4)^{1/L}$. Subtract this sum from κ_N and use $\|W^s - R_s\|_F^2 = \|W^s\|_F^2 - p_s^2$ to obtain (**??**). Alternatively, the product of the Frobenius energy terms is at least $(d/4)\prod_s \rho_s$ and at most $(\kappa_N/L)^L$, which proves the logarithmic bound. Finally use $e^u \ge 1 + u$ and $\log(1 + u) \le u$. □

Thus bounded excess permits only a bounded total squared residual after replacing internal matrices by rank-one matrices. This does not justify replacing those matrices in the computation: the suffix can amplify the replacement errors. The next example shows that even a constant excess allowance does not imply inverse-polynomial hidden-state robustness.

Proposition 1.2 (Bounded excess permits a fragile scalar transmission)**.** *For $d = 4$, every even $L \ge 4$, and every $N \ge 1$, there is a network with $z(x) = 1$ for all inputs and*

$$\kappa_N \le L + 76 \le K_*(4, L) + 76,$$

whose output can be made zero by a perturbation at one hidden layer of normalized RMS at most $L^{-1}e^{-\sqrt{L}/4}$. Here $K_*(4, L)$ is the globally minimum margin-$1/2$ energy characterized in the main text.

Proof. Put $n = L/2$, $h = L^{-1/2}$, $k_s = \min(s, L - s)$, and

$$r_s = \frac{e^{-h(k_s-1)}}{2k_s}, \qquad 1 \le s < L.$$

Give each first-layer row the sole constant-input weight $\operatorname{atanh}(1/2)$. Let every entry of W^s be b_s/N, where $b_s = \operatorname{atanh}(r_s)/r_{s-1}$, and let $v_i = 2/N$. The hidden vector is exactly $r_s\mathbf{1}$, so the output is $2r_{L-1} = 1$ and the energy is

$$4\operatorname{atanh}(1/2)^2 + \sum_{s=2}^{L-1} b_s^2 + 4.$$

For completeness, the following loose constants are uniform in depth. The ratios r_s/r_{s-1} occur in reciprocal pairs t_k, t_k^{-1}, where $t_k = (1+1/k)e^h$ and $1 \le k < n$. Since $h \le 1/2$,

$$t_k - t_k^{-1} \le e^h(2/k + 2h), \qquad t_k^2 + t_k^{-2} - 2 \le 8e^{2h}(k^{-2} + h^2).$$

Using $e^{2h} < 3$, $\sum_{k\ge1} k^{-2} < 2$, and $(n-1)h^2 \le 1/2$ gives $\sum_s (r_s/r_{s-1})^2 < L - 2 + 60$. Also $\max_s (r_s/r_{s-1})^2 < 12$ and $\sum_s r_s^2 < 1$. For $0 < t \le 1/2$ the power series gives $(\operatorname{atanh}(t)/t)^2 \le 1 + t^2$. Consequently

$$\sum_s b_s^2 < L - 2 + 72, \qquad 4\operatorname{atanh}(1/2)^2 \le 5/4, qquad \kappa_N < L + 76.$$

The baseline lower bound gives $K_*(4, L) \ge L$. Finally, adding $-r_n\mathbf{1}$ after the middle activation zeros that state and hence every later state. Its normalized RMS is

$$r_n = L^{-1}\exp\left(-\frac{L/2-1}{\sqrt{L}}\right) \le L^{-1}e^{-\sqrt{L}/4}.$$

□

This example represents a statistically easy target: the earlier theorem proves depth-independent strong-prior learning for constants. It therefore separates fragility of a cheap parameter realization from difficulty of learning its Boolean function. It does not show that the fragile realization is a minimum-energy representative.

Proposition 1.3 (Additional depth has asymptotically unit energy cost)**.** *Suppose a width-N, depth-L_0 network realizes f with margin at least $1/2$. Write $c^2 = N\|v\|_2^2 > 0$ and $E_{\text{pre}} = \kappa_N - c^2$. For every integer $t \ge 1$, there is a width-N, depth-$(L_0 + t)$ realization of the same function, with margin at least $1/2$ and energy at most*

$$E_{\text{pre}} + (t+13)\max\{c^2, 1\}^{1/(t+1)}.$$

Thus, for each fixed representable f, d, N, its strict-margin infimum energy satisfies $\kappa_{N,L}(f) = L + O_{f,d,N}(1)$ as $L \to \infty$.

Proof. Extend the scalar minimum $K_*(D, m)$ to any real $D > 0$: minimize $Da^2 + \sum_{j=2}^{m-1} b_j^2 + c'^2$ subject to $c' \tanh(b_{m-1}\tanh(\cdots\tanh a)) \ge 1/2$. The scalar baseline upper-bound argument works unchanged for real $D > 0$:

$$K_*(D, m) \le (m+12)(\max\{D, 4\}/4)^{1/m}.$$

Indeed, setting $q = (\max\{D,4\}/4)^{1/(2m)}$ and $r_j = q^j/[\sqrt{\max\{D,4\}}\min(j, m-j)]$ gives this bound by $a = \operatorname{atanh} r_1$, $b_j = \operatorname{atanh}(r_j)/r_{j-1}$, $c' = q$, and the summable reciprocal-ratio estimates used earlier.

Take the scalar tuple for $D = 4c^2$ and $m = t+1$. Replace the old readout by a hidden layer whose every row is $2av^T$, followed by the remaining scalar layers and readout. Its first new preactivation is $2az(x)$, whose absolute value is at least a and whose sign is $f(x)$. Monotonicity and oddness give the same sign with final margin at least $1/2$. The first new block has energy $4a^2c^2$; the remaining new blocks have exactly the remaining scalar costs. The total is therefore $E_{\text{pre}} + K_*(4c^2, t+1)$, proving the displayed upper bound. An arbitrarily small readout increase supplies strict margins. Expanding this bound as $t \to \infty$, and combining it with $L(d/4)^{1/L} \le \kappa_{N,L}(f)$, proves the final assertion. □

These results leave the unrestricted Gaussian sample-complexity question open. They rule out exponential growth of minimum target energy under depth padding, while showing that bounded excess does not make every cheap realization robust. A target-relative argument must also control competing functions and their prior masses; subtracting a common transmission cost alone does not do so.